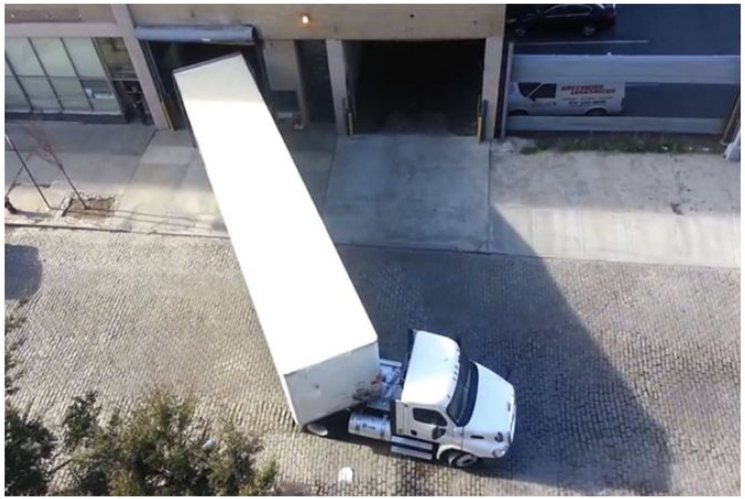
Trajectory-Aware RL for Autonomous Tractor-Trailer Reverse Docking

Nandan, Prince, Chris Pedretti

Task

Reverse a tractor with one semitrailer into a selected loading door in a simulated logistics yard.



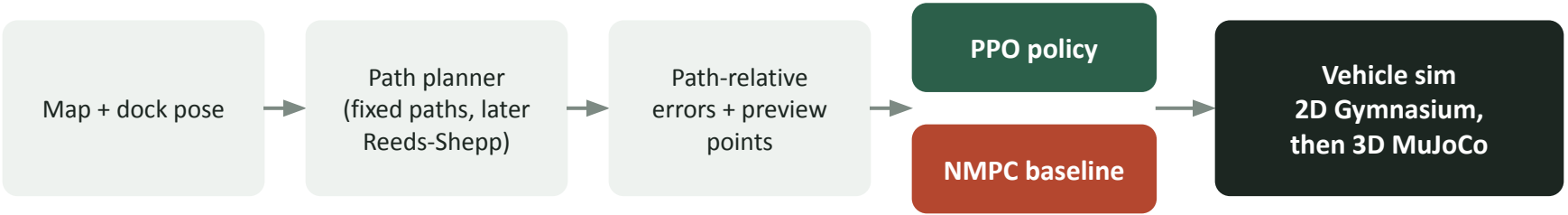
Objective

Test whether a PPO controller with path-relative inputs can dock reliably, and how it compares with classical model-based NMPC.

Success means

- Trailer within position and heading tolerance of the door
- No collisions with obstacles or the dock
- Articulation angle never exceeds the jackknife limit
- Target: ≥90% success on held-out nominal starts

Environment and method



Same interface for both controllers; vehicle state (pose, headings, articulation) fed back each step. Simulator ground truth gives localization.

Inputs

Trailer lateral error, tractor and trailer heading errors, articulation and steering angle, distance to goal, lateral and heading offsets of preview points

Output

Bounded steering-rate command at a fixed low reverse speed; a supervisor clips unsafe commands

Reward and training

Progress along path, tracking and articulation penalties, small steering-rate penalty, dock bonus, large collision or jackknife penalty. PPO (SB3) with domain randomization over start pose, trailer length, noise

Evaluation and baselines

Experiments

PPO	*vs*	**NMPC baseline**
Trajectory-relative obs	*vs*	**Global-coordinate obs**
Perfect state	*vs*	**Noisy state**

Fair comparison

- 100+ held-out trials per condition
- Identical start states and target docks for every controller
- Same vehicle model, limits, and supervisor
- Start pose and trailer geometry varied; failures categorized, not dropped

Metrics

- Success, collision, jackknife rate
- Final lateral, longitudinal, heading error
- Max articulation angle
- Maneuver time, path length
- Steering smoothness
- Controller compute time

Question: *does path-relative PPO match NMPC in nominal cases and hold up better under model mismatch and noise?*

Beyond Yan et al. (2024): adds a dock target, obstacles, an NMPC baseline, robustness tests, and MuJoCo transfer

Feasibility and plan

COMPLETED

- Problem definition and proposal
- Literature review
- Tool stack chosen: Gymnasium, MuJoCo, SB3, CasADi/IPOPT

PLANNED

1. 2D env, fixed paths, NMPC baseline
2. Train PPO: path-relative
3. Transfer to MuJoCo yard
4. Held-out evaluation, final report

COMPUTE

Simulation only, no hardware. 2D training runs on a laptop CPU; MuJoCo uses many parallel environments on CPU, with a single GPU optional.

Fallback (reduced scope)

Planner
fixed paths only

NMPC
pure pursuit or LQR

PPO
curriculum, report solved subset

MuJoCo
full study in 2D, 3D demo only

References

Yan, H., Zohdy, M. A., Alhawsawi, E., & Mahmoud, A. (2024). Trajectory state model-based reinforcement learning for truck-trailer reverse driving. ICRCA 2024. doi:10.1109/ICRCA60878.2024.10649133

Moradi, B., & Mehrandezh, M. (2022). A non-linear MPC local planner for tractor-trailer vehicles in forward and backward maneuvering. arXiv:2212.11427

Ma, H., Zhang, T., Li, N., Di Cairano, S., & Wang, Y. (2025). A hierarchical approach for tractor-trailer motion planning using graph search and reinforcement learning. ECC 2025. doi:10.23919/ECC65951.2025.11186959

Menz, C., & Wenzel, A. (2026). Edge-based imitation learning for autonomous tasks: A case study on tractor-trailer positioning. at – Automatisierungstechnik. doi:10.1515/auto-2025-0081